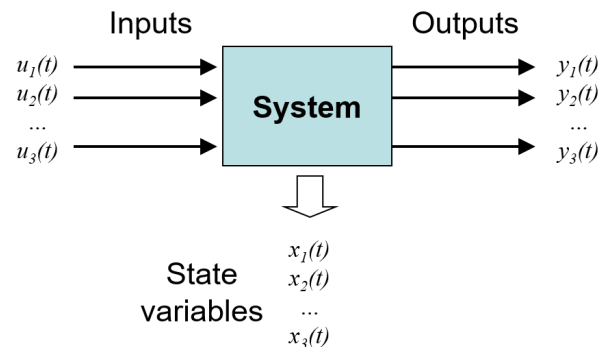


ESC-251 Systems Engineering

Systems

A system is a combination of interacting components that produces an output in response to an input. Systems can be mechanical, electrical, chemical, etc. The behavior of a system can be described using mathematical models, such as differential equations or transfer functions.



Static Systems

Output depends only on the current input and does not change with time, once the input stops, the output stops (ie. static displacement of a spring).

Dynamic Systems

Output depends on past and current inputs, and changes over time (ie. driven oscillators).

Parameter Variation

- **Time-varying:** coefficients of the system change with time (ie. changing mass or spring stiffness).
- **Time-invariant:** coefficients of the system do not change with time (ie. constant mass and spring stiffness).

For this course, we will focus on linear, time-invariant (LTI) systems. This means that the system's *equations of motion* are linear, and the coefficients do not change with time. A property of time-invariant systems is that a time delay in the input only delays the output by the same amount, with no dependence on the absolute time.

Linear systems also obey **superposition**, which means that its response to multiple inputs is the sum of its responses to each input *individually*.

Generally, our approach to systems analysis will be to model elements mathematically to derive the equations of motion. Then, we use the Laplace transform to the differential equations to either find the **transfer function** (the ratio of the output to the input in the Laplace domain s) or the response of the system in the time domain (solution of the differential equation).

Math Review

Complex Numbers

In systems we use $j = \sqrt{-1}$ instead of i to avoid confusion with current.

Complex Vector Representation

Rectangular form:

$$\vec{X} = a + jb$$

Polar form:

$$\vec{X} = M \angle \phi$$

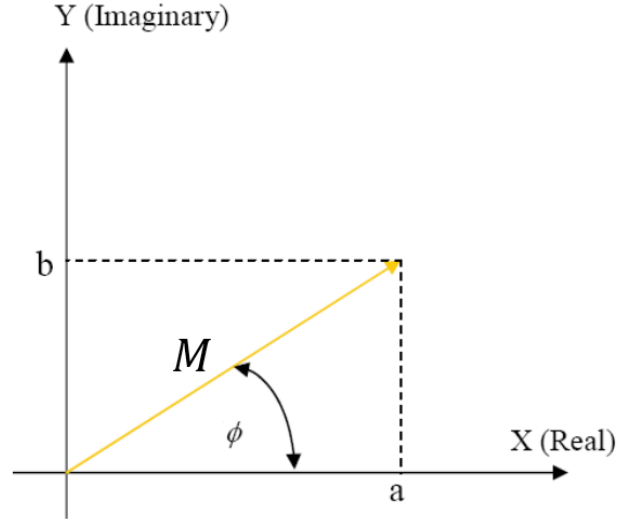
Complex Exponential form:

$$\vec{X} = M e^{j\phi}$$

Where:

$$M = \sqrt{a^2 + b^2}, \quad \phi = \arctan\left(\frac{b}{a}\right)$$

$$\implies a = M \cos(\phi), \quad b = M \sin(\phi)$$



Note: $\arctan$ has a range of $(-\pi/2, \pi/2)$, so we have to add or subtract π to get the correct angle based on the quadrant of the complex number.

Euler's Identity

$$e^{j\phi} = \cos(\phi) + j \sin(\phi)$$

Complex Conjugate

The complex conjugate of $\vec{X} = a + jb$ is $\vec{X}^* = a - jb$. In polar form, the complex conjugate is $M \angle -\phi$.

Operations

Addition/Subtraction:

$$(a + jb) + (c + jd) = (a + c) + j(b + d)$$

Multiplication/Division:

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc)$$
$$\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2}$$

Using Complex Exponential Form:

Multiplication:

$$(M_1 e^{j\phi_1}) (M_2 e^{j\phi_2}) = M_1 M_2 e^{j(\phi_1 + \phi_2)}$$

Division:

$$\frac{M_1 e^{j\phi_1}}{M_2 e^{j\phi_2}} = \frac{M_1}{M_2} e^{j(\phi_1 - \phi_2)}$$

Note: Multiplying by j is equal to *adding* a phase shift of 90° (or $\pi/2$ radians) in the complex plane.

$$j = e^{j\frac{\pi}{2}}$$

Also, if z and w are complex numbers, then:

$$|zw| = |z||w|$$

But

$$|z + w| \neq |z| + |w|$$

Differential Equations

Linear First Order Differential Equation

For a first-order linear differential equation of the form:

$$\frac{dy}{dx} + p(t)y = g(t)$$

Method:

Find the integrating factor:

$$\mu(t) = e^{\int p(t)dt}$$

The solution is given by:

$$y(t) = \frac{1}{\mu(t)} \int \mu(t)g(t)dt$$

Linear Second Order Differential Equation

For higher order homogeneous linear differential equations with constant coefficients of the form, we can use the characteristic equation to find the *homogeneous solution* (right hand side equals zero).

Method:

Given a homogeneous linear differential equation with constant coefficients:

$$ay'' + by' + cy = 0$$

We assume the solution of the form:

$$\begin{aligned} y(t) &= e^{rt} \\ y'(t) &= re^{rt} \\ y''(t) &= r^2 e^{rt} \end{aligned}$$

Then,

$$ar^2 + br + c = 0$$

We can solve for the roots r using the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: For higher order differential equations, we would get a polynomial of degree n , and we would solve for the roots by factoring.

Case 1: Real and Unique Roots

For roots r_1 and r_2 , the general solution is:

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

Case 2: Complex Roots

For roots $\alpha \pm \beta j$, the general solution is:

$$y(t) = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t$$

Case 3: Repeated Roots

For repeated root r , the general solution is:

$$y(t) = c_1 e^{rt} + c_2 t e^{rt}$$

Finally, we solve for the **particular solution** (non-zero input/forcing function) using the method of undetermined coefficients. We guess the form of the particular solution and solve for the coefficients by plugging it into the differential equation.

The **general solution** is the sum of the homogeneous and particular solutions:

$$y(t) = y_h(t) + y_p(t)$$

Laplace Transform

Laplace transform is the method we will mainly use to solve the differential equations of systems. It requires us to know the initial conditions of the system, but transfer functions (discussed later) assume *zero initial conditions*.

Method:

To solve a differential equation with given initial conditions using Laplace transforms:

1. Apply the Laplace transform $\mathcal{L}$ to both sides of the differential equation. This transforms the differential equation to an algebraic equation in the s -domain in terms of $Y(s) = \mathcal{L}\{y(t)\}$.
2. Solve the algebraic equation for $Y(s)$.
3. Apply the inverse Laplace transform $\mathcal{L}^{-1}$ to find $y(t)$, converting from the s -domain back to the time domain.

Typically, partial fraction decomposition is needed to find the inverse Laplace transform.

Laplace Transform Formulas (From Diff Eq. Notes)

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
$\mathcal{U}(t-a)f(t-a)$	$e^{-as}F(s)$
$e^{at}f(t)$	$F(s-a)$
$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
$\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$
$t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$
$f * g$	$F(s)G(s)$
$\delta(t-a)$	e^{-sa}
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$

Cover-Up/Limit Method

For solving partial fraction decomposition to find the inverse Laplace transform, we can use the cover-up/limit method to find the coefficients of the factors in the denominator.

- **Distinct Linear Factors:**

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

To find the coefficients, we use:

$$A = \lim_{x \rightarrow a} \left[(x-a) \frac{P(x)}{(x-a)(x-b)(x-c)} \right]$$

- “Cover up” factor in denominator
- Evaluate at the “root”

- **Ex:**

$$\frac{5x+3}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$$

$$A = \left. \frac{5x+3}{x-2} \right|_{x=1} = -8$$

$$B = \left. \frac{5x+3}{x-1} \right|_{x=2} = 13$$

- **Repeated Linear Factors (Derivative Limit Method):**

$$F(x) = \frac{P(x)}{G(x)(x-a)^n} = \underbrace{\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_n}{(x-a)^n}}_{\text{factors of } (x-a)^n} + \dots \text{ (terms for the factors of } G(x))$$

For the k^{th} coefficient:

$$A_k = \frac{1}{(n-k)!} \lim_{x \rightarrow a} \left\{ \frac{d^{n-k}}{dx^{n-k}} [(x-a)^n F(x)] \right\}$$

- When $k = n$ (the highest order factor), we have the same formula as for distinct linear factors.
- We use the derivative to “recover” the lower order roots.

- **Ex:**

$$F(x) = \frac{2x+3}{(x-1)^3(x+2)} = \frac{A_1}{(x-1)} + \frac{A_2}{(x-1)^2} + \frac{A_3}{(x-1)^3} + \frac{B}{x+2}$$

$$A_3 = \left. \frac{2x+3}{x+2} \right|_{x=1} = \frac{5}{3} \qquad A_1 = \left. \frac{1}{2!} \frac{d^2}{dx^2} \left(\frac{2x+3}{x+2} \right) \right|_{x=1} = -\frac{1}{27}$$

$$A_2 = \left. \frac{d}{dx} \left(\frac{2x+3}{x+2} \right) \right|_{x=1} = \frac{1}{9} \qquad B = \left. \frac{2x+3}{(x-1)^3} \right|_{x=-2} = -\frac{1}{27}$$

Note:

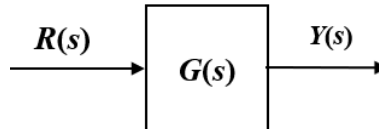
- For irreducible quadratic factors, this doesn’t apply (use $Cx + D$ and solve).
- Always check that (deg of num) < (deg of denom).

Response Analysis

Transfer Function

Transfer functions relate input to output in the Laplace domain, *assuming zero initial conditions*. It is defined as:

$$G(s) = \frac{Y(s)}{R(s)}$$



Transfer functions *don't capture the time domain response of the system*, because the input function is undefined. However, once we define an input function (step function, impulse, etc.), we can find the time domain response by taking the inverse Laplace transform of $Y(s) = G(s)R(s)$.

Since the transfer function assumes zero initial conditions, it captures only the **steady state response** of the system, which is the response due to the input function.

Note: When writing out the equations of motion (time domain), write it in the form

$$(\text{output}) = (\text{input})$$

$$(\text{dep. variables LHS}) = (\text{input RHS})$$

Transient vs Steady State Response

Transient Response is the part of the system's response that dies out over time, while **Steady State Response** is the part of the response that remains after the transient response has decayed.

Typically, the steady state response is due to an input to the system, this input can either be a forcing function (unit-impulse, unit-step, etc.) or some static (constant) input. The transient response is due to the initial conditions (x_0 , v_0 , etc.) of the system. Recall that the homogeneous solution corresponds to the transient response, while the particular solution corresponds to the steady state response.

Stability

In terms of natural (unforced/free) response,

- System is **stable** if $x(t) \rightarrow 0$ as $t \rightarrow \infty$
- System is **unstable** if $x(t) \rightarrow \infty$ as $t \rightarrow \infty$
- System is **marginally stable** if natural response neither decays nor grows but remains constant or oscillates

In terms of total response,

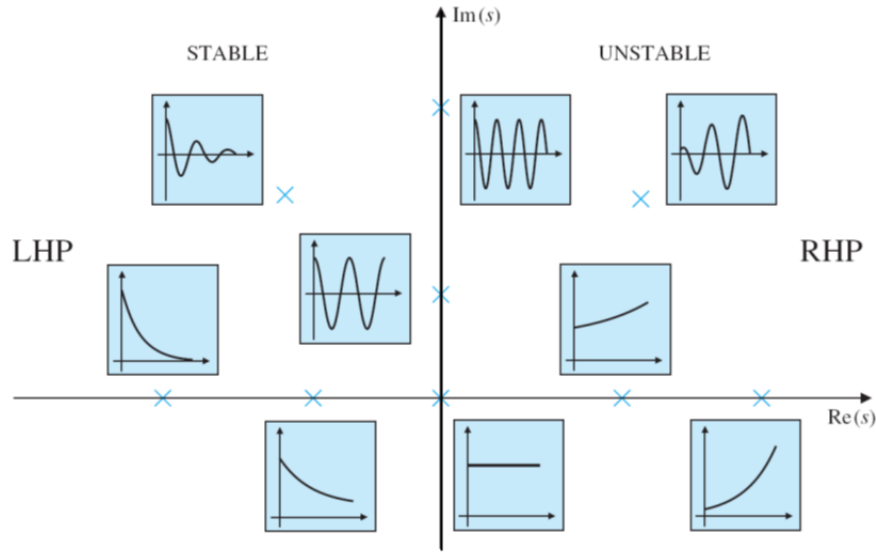
- System is **stable** if *every* bounded input yields a bounded output (**BIBO**)
- System is **unstable** if *any* bounded input yields an unbounded output

Mathematically, we can determine stability by looking at the roots of the characteristic equation which are the *poles* of the transfer function (a pole is a value of s that makes the denominator of the transfer function zero):

- Negative Real Part ($\text{Re}[s] < 0$) $\rightarrow$ **Stable**
- Positive Real Part ($\text{Re}[s] > 0$) $\rightarrow$ **Unstable**, grows exponentially
- One zero at $s = 0$ and one negative root $\rightarrow$ **Neutrally or marginally stable** (the negative root decays to zero, but the zero root stays constant, so its steady state response is marginally stable)
- Zero real part, non-zero imaginary $\rightarrow$ **Oscillatory response** (neutrally or marginally stable)

All roots must be stable for system to be considered stable! For example, it is possible to have stable and unstable oscillatory responses since the real part determines the amplitude of the oscillation and the imaginary part determines the frequency of the oscillation.

- Positive real part ($\text{Re}[s] > 0$), non-zero imaginary part ($\text{Im}[s] \neq 0$) $\rightarrow$ **Unstable**, flutter instability
- Negative real part ($\text{Re}[s] < 0$), non-zero imaginary part ($\text{Im}[s] \neq 0$) $\rightarrow$ **Stable**, damped oscillation



Note: The poles of the transfer function are the values that make the denominator zero, which are the roots of the characteristic equation. The location of the poles in the complex plane determines the stability of the system.

Identifying Frequency and Time Constants

Aside from determining the stability of a system based on its poles, we can also analyze the system's response in the time domain by looking at the time constant, natural frequency, and damping ratio of the system. These parameters can be identified from the transfer function or the characteristic equation of the system.

Time Constant

For a transfer function of a **first order system**:

$$G(s) = \frac{K}{\tau s + 1}$$

Where τ is the time constant and K is the steady state gain.

The time constant τ describes how quickly a system reaches steady state (shorter τ means faster response). For a first order system in the time domain, the time constant is the value for t such that $-Ct$ becomes -1 :

$$v(t) = v_0 e^{-Ct}, \quad \tau = \frac{1}{C}$$

At 1τ the system is 63.3% of the way to its final steady state value, at 2τ it is 86.5%, at 3τ it is 95%, and at 4τ it is 98.2%. Thus, we can say that the system has effectively reached its steady state value after 4τ .

The steady state gain K is defined as the steady state output of the system *given a unit step input*. Intuitively, it describes how much the system amplifies the input at steady state.

Natural Frequency

For a **second order system with no damping**, its characteristic equation in the time and laplace domains are as follows:

$$\ddot{x} + \omega_n^2 x = 0, \quad s^2 + \omega_n^2 = 0$$

Where ω_n is the natural frequency of the system.

Damping Ratio

For **second order damped systems**, we have a non-zero damping ratio ζ . For a second order system with damping, its characteristic equation is in the form:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0 \quad \text{or} \quad = \frac{f(t)}{m} \quad \text{if there is a forcing function}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad (\text{in laplace domain})$$

The damping ratio determines the nature of the system's response:

- $\zeta < 0$: Unstable system (response grows exponentially).
- $\zeta = 0$: Marginally stable (undamped/free oscillations at ω_n).
- $0 < \zeta < 1$: **Underdamped** (stable, oscillatory response with decaying amplitude).
- $\zeta = 1$: **Critically damped** (stable, fastest exponentially decaying return to equilibrium).
- $\zeta > 1$: **Overdamped** (stable, exponentially decaying response with slower return to equilibrium).

Note: The **time constant of a damped second order system** is given in terms of the *dominant pole* σ (since the system has 2 poles). The dominant pole is the pole closest to the imaginary axis (the slower pole), and the time constant is given by

$$\tau = \frac{1}{\sigma}$$

The faster pole has a shorter transient response and decays to zero much faster than the slower pole and thus the dominant pole determines the overall response of the system better.

Final/Initial Value Theorem

We can determine the initial and final values of a system's response by just looking at the transfer function without having to find the full time domain response.

Final Value Theorem

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

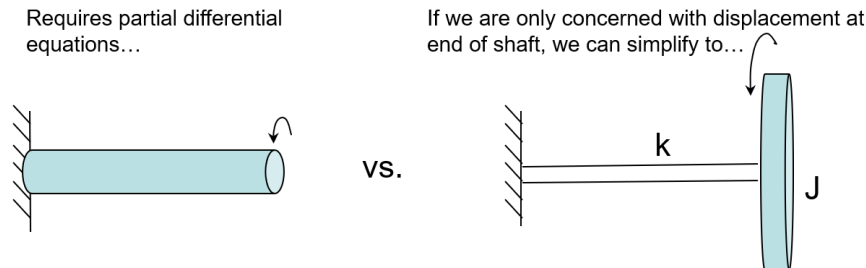
Initial Value Theorem

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

Mechanical Systems

Distributed vs Lumped Systems

Lumped systems are systems where the properties (mass, stiffness, damping) are concentrated at discrete points, these systems are easier to work with since we can define a finite number of degrees of freedom. Distributed systems have properties that are distributed continuously along the system, these systems are more complex and require partial differential equations to model.



Mechanical Elements

Inertia/Mass Elements

The inertia elements of a mechanical system are the mass and moment of inertia that appear in a Newton's second law equation.

$$\sum F = ma, \quad \sum \tau = J\alpha$$

Inertia elements also store energy in the system as kinetic or potential energy. The kinetic energy of a mass is given by

$$T_{\text{trans}} = \frac{1}{2}m\dot{x}^2 \quad \text{or} \quad T_{\text{rot}} = \frac{1}{2}J\dot{\theta}^2$$

Spring/Stiffness Elements

Spring elements provide a restoring force that is proportional to the displacement of the system. The force exerted by a spring is given by Hooke's law:

$$F = kx \quad \text{or} \quad \tau = k_t\theta \quad \text{for a torsional spring}$$

and store potential energy in the system:

$$U = \frac{1}{2}kx^2 \quad \text{or} \quad U = \frac{1}{2}k_t\theta^2$$

It obeys the following assumptions:

- Lumped parameter (spring is massless and has no damping)
- Perfectly elastic (no energy loss)

Damping/Dissipative Elements

Damping elements provide a resistive force that is proportional to the velocity of the system. The force exerted by a damper is given by:

$$F = b\dot{x}$$

It obeys the following assumptions:

- No mass
- No energy storage (no stiffness)
- Is linear (force is proportional to velocity)

Power

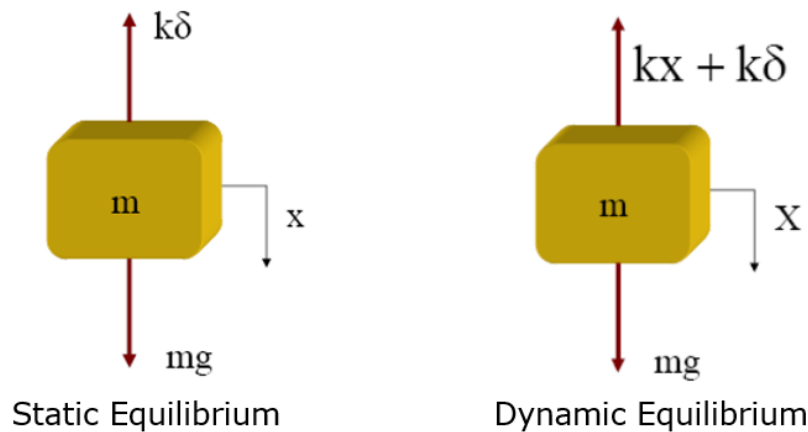
Power = effort $\times$ flow. In mechanical systems, effort is force and flow is velocity, so power is given by:

$$P = F \cdot v$$

In terms of rotation:

$$P = \tau \cdot \omega$$

Spring-Mass System (1 DOF)



An important concept for a spring-mass system is the *static deflection* δ , defined as the displacement of the spring under the weight of the mass. If we choose the coordinate x to measure displacement from the static equilibrium position, gravity does not need to appear in the equation of motion because the weight is exactly balanced by the spring force at equilibrium, where $mg = k\delta$. The equation of motion for the system therefore becomes:

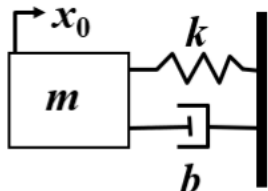
$$m\ddot{x} + kx = 0$$

Note: We can also ignore gravitational potential energy if we choose the x coordinate to measure displacement from the static equilibrium position, because the potential energy terms simplify to

$$U = \frac{1}{2}kx^2 + \text{constant}$$

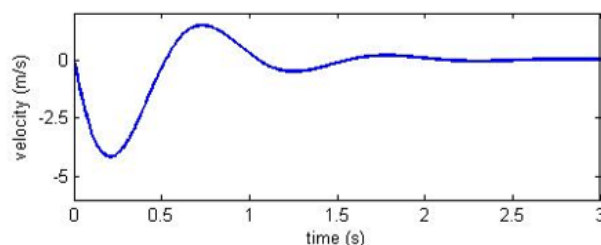
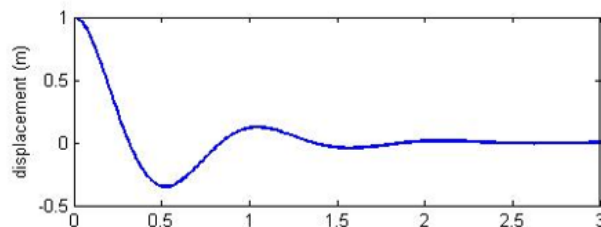
However, for more complex systems, we will need to first draw the static equilibrium FBD to determine whether we can ignore gravity in the equations of motion.

Response Due to Initial Conditions Example

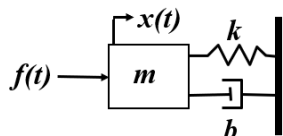


$$m\ddot{x} + b\dot{x} + kx = 0$$

Notice that the response reaches steady state at $x = 0$ and $v = 0$ because there is no forcing function, so the system is stable. The initial displacement was what generated the transient response.

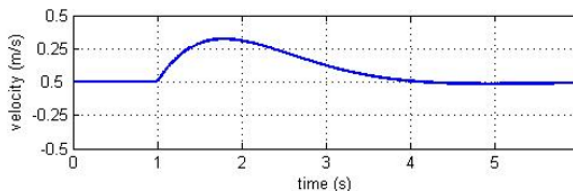
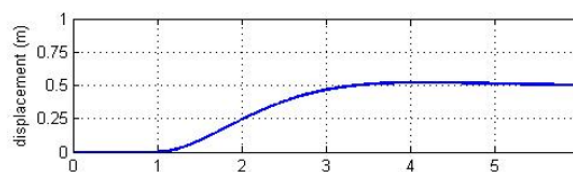
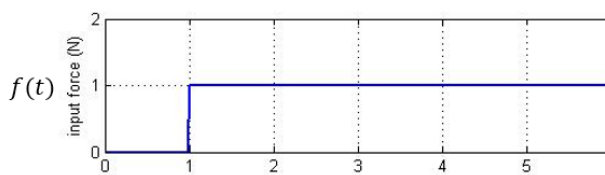


Step Response Example



$$m\ddot{x} + b\dot{x} + kx = f(t)$$

Notice that the response is also delayed by 1 second since the input is a step function that starts at $t = 1$. Now, the steady state response is non-zero because there is a forcing function, but the transient response still decays to zero over time.



The solution via Laplace transforms is (where $1(t - 1)$ is the **unit step function**):

$$x(t) = \frac{1}{2} \cdot 1(t - 1) - \frac{1}{2}e^{-(t-1)} [\cos(t - 1) + \sin(t - 1)] \cdot 1(t - 1)$$

Recall: The unit step function or Heaviside function is defined as

$$1(t - a) = \begin{cases} 0 & 0 \leq t < a \\ 1 & t \geq a \end{cases}$$

Using this function, we can represent piecewise functions as a single function. For example,

$$f(t) = \begin{cases} g(t) & 0 \leq t < a \\ h(t) & t \geq a \end{cases}$$

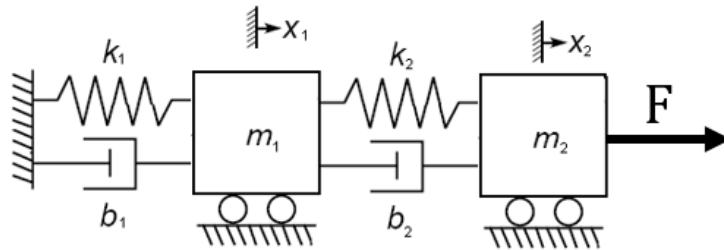
We can define this in the following way

$$f(t) = g(t)(1 - 1(t - a)) + h(t)1(t - a)$$

We can do this using the fact that $1(t - a)$ is 0 before $t = a$ and 1 after $t = a$. Thus, $1 - 1(t - a)$ is 1 before $t = a$ and 0 after $t = a$.

Two DOF Systems

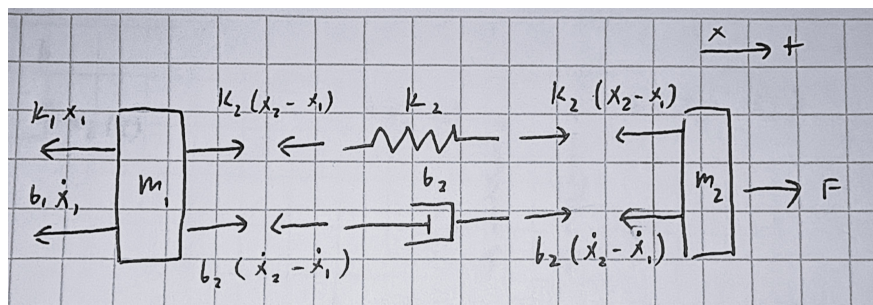
A degree of freedom (DOF) is an independent coordinate that can be used to describe the configuration of a system. What makes a 2 DOF system different from a 1 DOF system is that the equations of motion are coupled, meaning that the motion of one mass affects the motion of the other mass.



To solve for the equations of motion, we start with an assumption of the relative positions x_1 and x_2 of the two masses. This way we can know what direction the spring forces are acting in. Note that we are measuring x_1 and x_2 relative to their static deflections.

Assumptions: $x_2 > x_1$, $\dot{x}_2 > \dot{x}_1$

Now we can draw the FBDs, keeping in mind that Newton's third law applies to the spring forces in between the two masses.



Equations of motion:

$$\sum F_1 = m_1 \ddot{x}_1 = -k_1 x_1 - b_1 \dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1)$$

$$\sum F_2 = m_2 \ddot{x}_2 = F(t) - k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1)$$

Now we can apply the Laplace transform to both equations, but since they are coupled, we will have to solve the system of equations to find the transfer function for each mass (substitute one equation into the other to eliminate one of the variables).

Energy Methods

Instead of using Newton's second law to derive the equations of motion, we can also use energy methods, which are often easier for more complex systems. The method is as follows:

- Define the total kinetic energy T and potential energy U of the system.
- Because energy is conserved, we can differentiate the total energy to find the equations of motion.

$$\boxed{\frac{d(T + U)}{dt} = 0}$$

Note that this only works for conservative systems, *where there is no energy loss due to damping or friction*. For non-conservative systems, it is easier to use Newton's second law to derive the equations of motion.

Rolling Without Slipping

Recall that when an object rolls without slipping, it obeys these conditions:

- $v_{\text{CM}} = R\omega$
- $a_{\text{CM}} = R\alpha$
- $v_{\text{contact}} = 0$

Note: Don't forget to include both translational and rotational kinetic energy, as well as gravitational potential energy if the system is not at static equilibrium.

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J\dot{\theta}^2$$

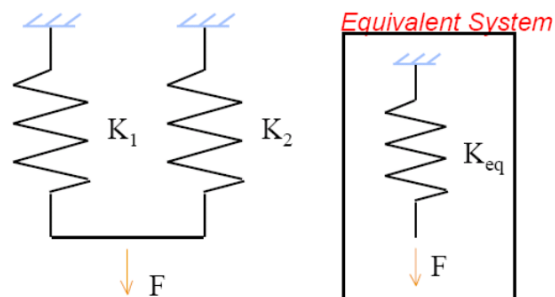
Sometimes when using small angle approx, we will need to use higher order approximations to not have a zero after differentiating.

Combining Stiffness

Springs in Parallel

When two springs are in parallel, they see the *same deflection* but feel different forces (if $k_1 \neq k_2$). The equivalent stiffness is the sum of the two stiffnesses:

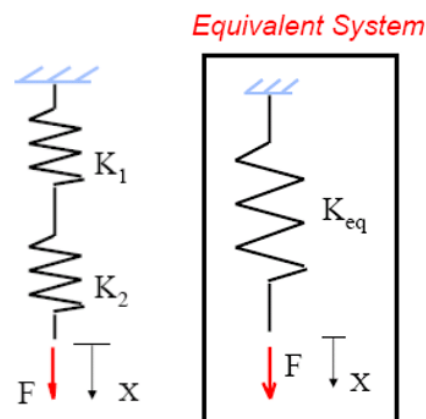
$$k_{eq} = k_1 + k_2$$



Springs in Series

When two springs are in series, they see the *same force* but feel different deflections (if $k_1 \neq k_2$). The equivalent stiffness is the reciprocal of the sum of the reciprocals of the two stiffnesses:

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$



Stiffness Equivalent: Buoyancy

Since the buoyant force is given as:

$$F_b = \rho g A x$$

Where x is the depth of the object in the fluid, A is the cross-sectional area of the object, and ρ is the density of the fluid, we can treat the buoyant force as a spring force with an equivalent stiffness of:

$$k_{eq} = \rho g A$$

Common Moments of Inertia

- **Point mass** (mass m at distance r from axis of rotation)

$$I = mr^2$$

- **Thin rod** (length L , mass m)

$$I = \frac{1}{12}mL^2 \quad (\text{about center, axis perpendicular to rod})$$

$$I = \frac{1}{3}mL^2 \quad (\text{about one end, axis perpendicular to rod})$$

- **Solid disk** (radius R , mass m)

$$I = \frac{1}{2}mR^2 \quad (\text{about center, axis perpendicular to plane})$$

- **Thin hoop / ring** (radius R , mass m)

$$I = mR^2 \quad (\text{about center, axis perpendicular to plane})$$

- **Solid cylinder** (radius R , mass m)

$$I = \frac{1}{2}mR^2 \quad (\text{about central symmetry axis})$$

- **Solid sphere** (radius R , mass m)

$$I = \frac{2}{5}mR^2 \quad (\text{about any diameter through center})$$

- **Thin spherical shell** (radius R , mass m)

$$I = \frac{2}{3}mR^2 \quad (\text{about any diameter through center})$$

- **Rectangular plate** ($a \times b$, mass m)

$$I = \frac{1}{12}m(a^2 + b^2) \quad (\text{about center, axis perpendicular to plate})$$

Electrical and Electro-Mechanical Systems

We will begin with electrical systems (circuits), and then move on to electro-mechanical systems, which contain both electrical and mechanical elements. Typically, electro-mechanical systems describe a *linear* relationship between electrical inputs (voltage or current) and mechanical outputs (velocity, torque, etc.). This will result in a multiple degree of freedom system and will be solved using the same methods as mechanical systems (substitution).

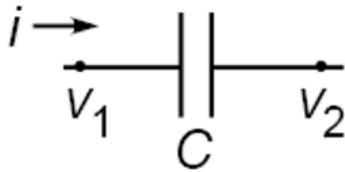
Electrical Elements

- Resistor



$$\begin{aligned}v_R &= iR \\ i(t) &= \frac{1}{R}(v_1 - v_2) \\ I(s) &= \frac{1}{R}(V_1 - V_2)\end{aligned}$$

- Capacitor

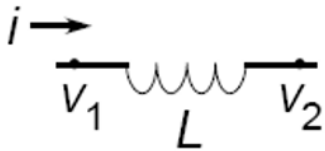


$$\begin{aligned}q &= Cv_c \\ \frac{d}{dt}(q) &= \frac{d}{dt}(Cv_c) \implies \boxed{i(t) = C \frac{dv_C}{dt}} \\ I(s) &= Cs(V_1 - V_2)\end{aligned}$$

$I(s)$ comes from the Laplace transform of $i(t)$:

$$i(t) = C \frac{d}{dt} \underbrace{(V_1 - V_2)}_{v_c}$$

- Inductor



$$\begin{aligned}v_L &= L \frac{di}{dt} \\ i(t) &= i_o + \frac{1}{L} \int v_L(t) dt, \text{ zero IC} \\ I(s) &= \frac{1}{Ls}(V_1 - V_2)\end{aligned}$$

To write the equations of motion (characteristic equation) for an electrical system, we use Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL) (see ESC 221 notes).

Impedance

Impedance Z is defined as the ratio of the voltage across an electrical element to the current through the element in the Laplace domain. Using impedance, every electrical element can be treated as a resistor, following an Ohm's law-like relationship:

$$Z = \frac{V(s)}{I(s)} \implies i = \frac{v_1 - v_2}{Z}, \quad v_1 - v_2 = iZ$$

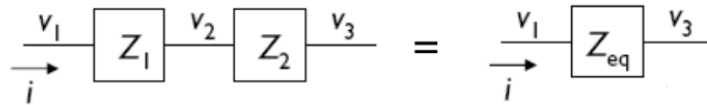
The impedance of each element is as follows:

- Resistor: $Z_R = R$
- Capacitor: $Z_C = \frac{1}{Cs}$
- Inductor: $Z_L = Ls$

Summing Impedances

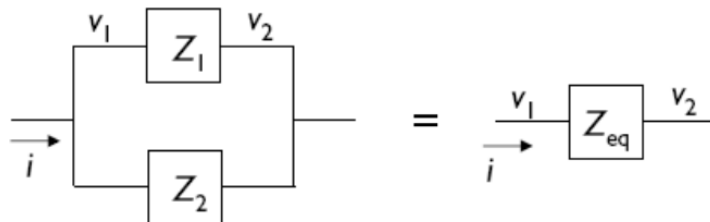
Since we can treat every element as a resistor, we can sum impedances in series and parallel just like we would for resistors:

- **Series Impedances**



$$Z_{eq} = Z_1 + Z_2$$

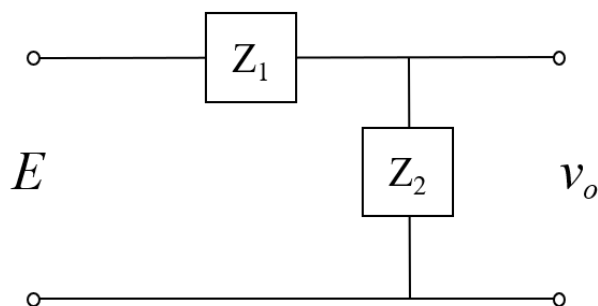
- **Parallel Impedances**



$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} \implies Z_{eq} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Impedance Voltage Division

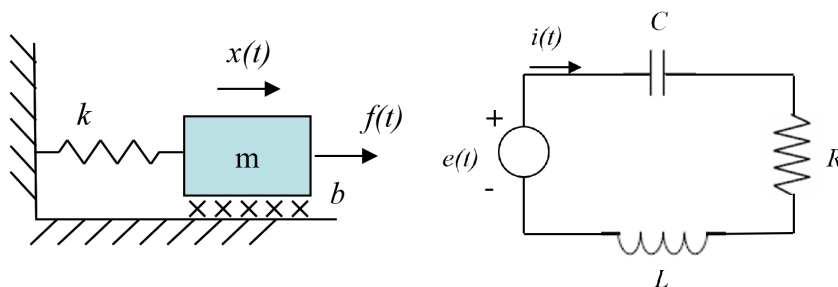
Like the voltage division rule for resistors, we can also use the impedance to find the voltage across an element in a series circuit:



$$v_o = \frac{Z_2}{Z_1 + Z_2} E$$

Note: When writing the characteristic equation for an electrical system using impedances, we are in the Laplace domain, *not the time domain!* This is often easier, though nodal and mesh analysis can also be used to write the equations of motion in the time domain.

Electrical and Mechanical Analogous Systems



Looking at the characteristic equations for the damped mass-spring system and the RLC circuit, we can see that they have the same form:

$$M \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = f(t)$$

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = e(t)$$

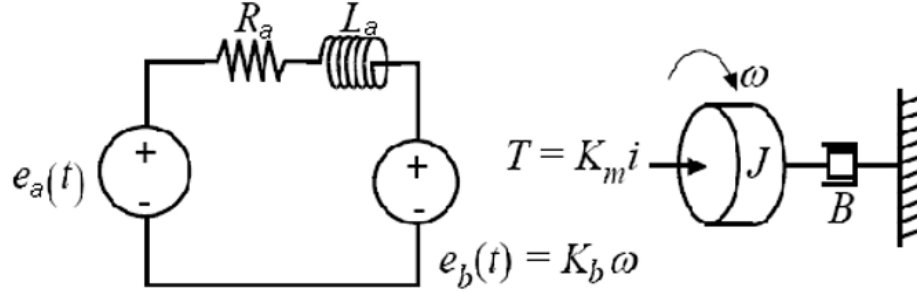
Power

Like mechanical systems, we can also define power for electrical systems as the product of effort $\times$ flow:

$$\begin{aligned} P &= v \cdot i \\ &= \tau \cdot \omega \quad (\text{for rotational mechanical systems}) \\ &= F \cdot v \quad (\text{for translational mechanical systems}) \end{aligned}$$

Electromechanical Systems (DC Motor)

The most common type of electromechanical system is a DC motor, which converts current to torque. It is commonly modeled as follows:



Where:

R_a = armature resistance (Ω)

L_a = armature inductance (H)

i_a = armature current (A)

e_a = applied armature voltage (V)

e_b = back emf (voltage produced by rotating armature) (V)

θ = angular displacement of motor shaft (rad)

ω = angular velocity of motor shaft (rad/s)

T = torque developed by motor (N-m)

J = moment of inertia of motor (kg-m^2)

B = viscous friction coefficient of motor load (N-s/m or kg/s)

K_m = motor torque constant (relates torque produced by electric part)

K_b = back emf or generator constant (relates voltage induced by motion)

The inertia term J describes the inertia of the motor itself, typically if we are modeling a motor connected to some additional load, we will need to add the inertia of the load to J (if it is a *rigid* connection to the motor shaft). However, the load can be connected via a gear train or a flexible shaft, in which case we will need to draw out the FBDs to determine the equations of motion.

Like mentioned before, this is a 2 DOF system since the electrical and mechanical parts are coupled, so we will have to perform substitutions to find the transfer functions.

$$e_a - R_a i - L_a \frac{di}{dt} - K_b \omega = 0 \quad (\text{KVL})$$

$$\sum \tau = J \frac{d\omega}{dt} = K_m i - B\omega \quad (\text{Newton})$$

$$\Rightarrow \frac{\Omega(s)}{E_a(s)} = \frac{K_m}{(L_a s + R_a)(J s + B) + K_m K_b} \quad (\text{Transfer Function})$$

DC Motor Specs & Characterization

- **DC Gain (K)**

By definition, the DC gain is the gain of the system at *steady state*. Therefore, time varying terms will go to zero:

$$\frac{di}{dt} = 0, \quad \frac{d\omega}{dt} = 0$$

In the transfer function (Laplace domain), this corresponds to $s = 0$ (essentially just applying the final value theorem for a unit step input):

$$K = \frac{K_m}{R_a B + K_m K_b}$$

Finally, by setting $B = 0$, we find the ideal max DC gain of the motor without any load/friction:

$$K = \frac{1}{K_b}$$

- **Electrical Time Constant (τ_{elec})**

Suppose we apply a *brake* to the motor shaft so that $\omega = 0$ and thus $e_b = 0$, then the electrical time constant describes how fast the current (and thus torque by $T = K_m i$) reaches steady state. Rewriting the KVL equation with $\omega = 0$ gives:

$$\begin{aligned} e_a - R_a i - L_a \frac{di}{dt} &= 0, \quad i = \frac{T}{K_m} \\ \implies e_a - R_a \frac{T}{K_m} - L_a \frac{d}{dt} \left(\frac{T}{K_m} \right) &= 0 \\ \implies \frac{T(s)}{E_a(s)} &= \frac{K_m}{L_a s + R_a} \\ \tau_{\text{elec}} &= \frac{L_a}{R_a} \end{aligned}$$

- **Mechanical Time Constant (τ_{mech})**

Suppose the motor is allowed to freely spin down from some initial speed with zero applied voltage (free response) and set $B = 0$, $L = 0$. Rewriting the transfer function gives:

$$\begin{aligned} \frac{\Omega(s)}{E_a(s)} &= \frac{K_m}{R_a J s + K_m K_b} \\ \tau_{\text{mech}} &= \frac{R_a J}{K_m K_b} \end{aligned}$$

This describes how fast an applied torque (or lack of) changes the speed of the motor, in terms of inertia and friction.

Note: If we substitute known values into the general $\Omega(s)/E_a(s)$ transfer function, the slower pole corresponds to the mechanical time constant and the faster pole corresponds to the electrical time constant. This is because the electrical response results in a much faster transient response that decays to zero.

Rewriting the KVL equation at steady state ($di/dt = 0$, $d\omega/dt = 0$) gives us a way to relate the torque and applied voltage at steady state. Using this, we can determine the no load speed and stall torque of the motor.

$$e_a - R_a i - K_b \omega = 0, \quad i = \frac{T}{K_m}$$

$$\implies e_a - R_a \frac{T}{K_m} - K_b \omega = 0$$

- **No Load Speed** ($T = 0$)

The no load speed is the max speed (ω) of the motor such that the back emf e_b is equal to the applied voltage e_a , which means that there is no current flowing through the motor and thus no torque produced. Setting $T = 0$ gives:

$$\omega_{\text{no load}} = \frac{e_a}{K_b}$$

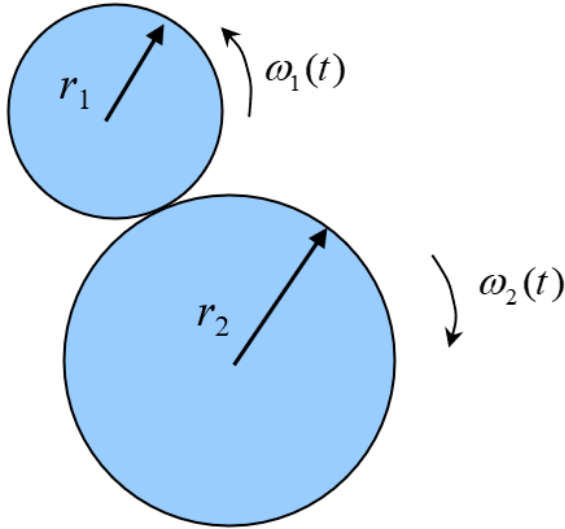
- **Stall Torque** ($\omega = 0$)

Stall torque is the max torque (T) generated by the motor when it is stopped ($\omega = 0$), thus there is no back emf and the current is at its max. Setting $\omega = 0$ gives:

$$T_{\text{stall}} = \frac{K_m}{R_a} e_a$$

Note: These formulas are NOT general, they only apply to the specific DC motor model shown above. If we have a different set-up with gears, shafts, or a load, then we will need to repeat these processes with the new equations of motion to characterize the DC motor system.

Gears



Gear ratios can be defined in terms of the number of teeth on each gear or the radius of each gear:

$$\text{Gear ratio} = \frac{N_2}{N_1} = \frac{r_2}{r_1}$$

Where N is the number of teeth and r is the radius of the gear. They mean the same thing since the number of teeth is proportional to the radius of the gear (same tooth size/module).

Since the gears rotate without slipping, the relative velocity at the contact point is zero, following the sign convention in the diagram above, we can say that

$$r_1\omega_1 = r_2\omega_2 \quad \implies \quad \frac{\omega_2}{\omega_1} = \frac{r_1}{r_2} = \frac{N_1}{N_2}$$

Neglecting friction loss, the torque is also related by the gear ratio:

$$\tau_1\omega_1 = \tau_2\omega_2 \quad \implies \quad \frac{\tau_2}{\tau_1} = \frac{\omega_1}{\omega_2} = \frac{N_2}{N_1}$$

Note: It is best to draw the internal reaction forces between the gears to determine the direction of the torques in the FBD and resulting equations of motion for each gear.

State Space Representation

We are able to rewrite the equations of motion of a system in the time domain into a matrix form using state variables. This is useful because it allows us to write an n-th order system as a system of n first order equations.

- **State Variables**

State variables are a set of variables that completely describes the behavior of the system along with the input.

- **State Vector**

The state vector is a column vector containing all the state variables.

The state space representation of a system is given by the following form:

$$\begin{aligned}\dot{\underline{x}} &= [A]\underline{x} + [B]\underline{u} \\ \underline{y} &= [C]\underline{x} + [D]\underline{u}\end{aligned}$$

Where:

- $\underline{x}$ = state vector
- $\underline{u}$ = input vector
- $\underline{y}$ = output vector
- $[A]$ = state matrix
- $[B]$ = input matrix
- $[C]$ = output matrix
- $[D]$ = direct transition matrix (output related to input)

Starting from the equations of motion and the defined state variables, we can rewrite the equations of motion in terms of the state variables. Ensure that the matrix multiplication (dot product between rows and columns) correctly reproduces the rewritten equations of motion and that the dimensions are consistent.

The output matrix is often either a matrix that selects certain state variables as outputs, or a matrix that operates on the state variables to produce the output (ie. momentum is a velocity state variable multiplied by mass). The direct transmission matrix is often zero, but it can be non-zero if the output is directly related to the input (ie. if the output is the input voltage in an electrical system).

Transfer Function Matrices

Like how we can derive transfer functions in the Laplace domain from the equations of motion, we can do the same in state space representation using matrix arithmetic. Starting with:

$$\begin{aligned}\dot{\underline{x}} &= [A]\underline{x} + [B]\underline{u} \\ \underline{y} &= [C]\underline{x} + [D]\underline{u}\end{aligned}$$

Taking the Laplace transform:

$$\begin{aligned}s\underline{X} &= [A]\underline{X} + [B]\underline{U} \\ \underline{Y} &= [C]\underline{X} + [D]\underline{U}\end{aligned}$$

Isolate the State Vector ($\underline{X}$):

$$\underline{X} = [sI - A]^{-1}[B]\underline{U}$$

Substitute $\underline{X}$ into the Output Equation:

$$\underline{Y} = [C] \underbrace{([sI - A]^{-1}[B]\underline{U})}_{\text{Sub for } \underline{X}} + [D]\underline{U}$$

Factor out $\underline{U}$ for the Transfer Function:

$$\boxed{\frac{\underline{Y}}{\underline{U}} = [C][sI - A]^{-1}[B] + [D]}$$

Recall: We can invert a 2x2 matrix using the formula:

$$\begin{aligned}A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ A^{-1} &= \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \text{if } ad - bc \neq 0.\end{aligned}$$